

† Singular value decomposition.

Let V, W be inner product space and let $T: V \rightarrow W$ be a linear.

Then, there exists orthonormal basis $\{v_1, \dots, v_n\}$ and $\{u_1, \dots, u_n\}$ of V, W respectively and positive scalars $\sigma_1 \geq \dots \geq \sigma_r$ s.t

$$T(v_i) = \begin{cases} \sigma_i u_i & (1 \leq i \leq \text{rank } T) \\ 0 & (i > \text{rank } T) \end{cases}$$

Conversely,

proof. Let $x \in \ker(T^*T)$. $(T^*T)(x) = 0$ implies

$$0 = \langle 0, x \rangle = \langle (T^*T)x, x \rangle = \langle Tx, Tx \rangle \Rightarrow T(x) = 0.$$

Therefore, $\ker T^*T \subseteq \ker T$, and the reverse inclusion is clear.

By the Dimension Thm, $\text{rank } T = \text{rank } T^*T$. Moreover, for any non-zero $x \in V$,

$$\langle T^*T(x), x \rangle = \langle T(x), T(x) \rangle = \|T(x)\|^2 \geq 0$$

Thus T^*T is positive semidefinite. Equivalently every eigenvalue of T^*T is non-negative and has an orthonormal basis of eigenvectors. Let $r = \text{rank } T$. Then,

denote the positive eigenvalues as

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_r > 0$$

and corresponding eigenvectors as

$$\{v_1, \dots, v_r, \dots, v_n\}$$

For each i , define $\sigma_i = \sqrt{\lambda_i}$ and $u_i = \frac{1}{\sigma_i} T(v_i)$. Then for any $1 \leq i, j \leq r$,

$$\langle u_i, u_j \rangle = \left\langle \frac{1}{\sigma_i} T(v_i), \frac{1}{\sigma_j} T(v_j) \right\rangle = \frac{1}{\sigma_i \sigma_j} \langle T^*T(v_i), v_j \rangle = \frac{\sigma_i^2}{\sigma_i \sigma_j} \langle v_i, v_j \rangle = \delta_{i,j}.$$

Therefore $\{u_1, \dots, u_r\}$ is orthonormal. Extending this to $\{u_1, \dots, u_n\}$, orthonormal basis for W .

And $T^*T(v_i) = 0 \Rightarrow T(v_i) = 0$ was proven.

⌈ Singular Value Decomposition.

Let $A \in M_{m,n}$ with $\text{rank } A = r$, with singular values $\sigma_1 \geq \dots \geq \sigma_r$.

Define a Matrix in $M_{m,n}$ as

$$\Sigma = \begin{cases} \sigma_i & \text{if } i \leq r \\ 0 & \text{otherwise} \end{cases}$$

Then, there exists unitary $U \in M_{m,m}$ and unitary $V \in M_{n,n}$ s.t.

$$A = U \Sigma V^*$$

$$U = \begin{pmatrix} u_1 & \dots & u_m \end{pmatrix}, \quad V = \begin{pmatrix} v_1 & \dots & v_n \end{pmatrix}$$

⌈ The Polar Decomposition.

For any $A \in M_{n,n}$, there exists unitary W and $P \geq 0$ s.t.

$$A = WP$$

Furthermore, if A is invertible, this representation is unique.

Proof. By the SVD, there exists unitary U, V and diagonal $\Sigma \geq 0$ s.t.

$$A = U \Sigma V = \underbrace{UV}_W \underbrace{V^* \Sigma V}_P$$

To show uniqueness, suppose A is invertible and

$$A = W_1 P_1 = W_2 P_2 \quad W_1, W_2 \text{ unitary, } P_1, P_2 \geq 0$$

Then, $W_2^* W_1 = P_2 P_1^{-1}$ unitary,

$$I = (P_2 P_1^{-1})^* (P_2 P_1^{-1}) = P_1^{-1} P_2^2 P_1^{-1} \Rightarrow P_1^2 = P_2^2 \Rightarrow P_1 = P_2.$$

